Studybeesofficial
EDUCATIONAL PLATFORM
www.studybeesofficial.com

SSC-I NATIONAL BOOK FOUNDATION FBISE

MATHEMATICS TEST - CHAPTER 6: TRIGONOMETRY AND BEARING

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. The radian measure system is also called the	English system	French system	Sexagesimal system	Circular system	○○○●
2. To convert degrees into radians, multiply by	180/π	π/360	π/180	360/π	○○●○
3. The area of a sector is given by	$A = r^2\theta$	$A = r\theta$	$A = \frac{1}{2}r^2\theta$	$A = \frac{1}{2}r\theta$	○○●○
4. A quadrantal angle is a multiple of	45°	60°	30°	90°	○○○●
5. Arc length and the radian measure of its central angle are	Directly proportional	Inversely proportional	Equal	Unrelated	●○○○
6. Two angles in standard position with the same terminal ray are called	Coterminal angles	Complementary angles	Supplementary angles	Adjacent angles	●○○○
7. One radian is the central angle of an arc whose length equals	The diameter	The circumference	Half the radius	The radius of the circle	○○○●
8. One degree equals	100 minutes	3600 minutes	60 minutes	30 minutes	○○●○
9. The sexagesimal system is also called the	English system	French system	Circular system	Metric system	●○○○
10. The equation of a unit circle centred at the origin is	$x^2 - y^2 = 1$	$x^2 + y^2 = 1$	$x + y = 1$	$x^2 + y^2 = 2$	○●○○
11. An angle is negative if it is measured in the	Clockwise direction	Anti-clockwise direction	Upward direction	Zero direction	●○○○
12. To convert radians into degrees, multiply by	180/π	π/180	360/π	π/360	●○○○
13. The ratio of arc length to radius is called	Degree	Radian	Minute	Second	○●○○
14. An angle is in standard position if its vertex is at the	Point (1, 0)	Origin	Point (0, 1)	Any point	○●○○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
15. The central angle of each of 360 equal arcs of a circle is	1 degree	1 minute	1 radian	1 second	●○○○
16. To convert seconds into minutes, we	Multiply by 60	Multiply by 3600	Divide by 60	Divide by 100	○○●○
17. In radians, quadrantal angles are multiples of	π/4	π/2	π/3	π/6	○●○○
18. The centesimal system is also called the	English system	Circular system	Radian system	French system	○○○●
19. One minute equals	60 seconds	100 seconds	3600 seconds	30 seconds	●○○○
20. A complete angle in the circular system measures	π radians	π/2 radians	2π radians	4π radians	○○●○
21. Two angles are coterminal if their measures differ by a multiple of	90°	180°	360°	45°	○○●○
22. π radians equal	90°	360°	270°	180°	○○○●
23. In standard position, the initial arm lies along the	Negative x-axis	Positive y-axis	Positive x-axis	Negative y-axis	○○●○
24. In the sexagesimal system, one complete circle is divided into	180 equal arcs	100 equal arcs	400 equal arcs	360 equal arcs	○○○●
25. Arc length for r = 10 cm and θ = π/5 is	2π cm	10π cm	4π cm	3π cm	●○○○
26. Arc length for r = 5 cm and θ = π/3 is	10π/3 cm	8π/3 cm	5π/6 cm	5π/3 cm	○○○●
27. Sector area for r = 12 cm and θ = π/4 is	19π cm²	18π cm²	9π cm²	3π cm²	○●○○
28. 20° in minutes is	120′	1200′	20′	2000′	○●○○
29. The angle 405° is coterminal with	495°	225°	135°	45°	○○○●
30. Arc length for r = 4 cm and θ = π/2 is	4π cm	π cm	2π cm	3π cm	○○●○
31. For r = 3 cm and θ = 4 rad, area of sector =	18 cm²	9 cm²	20 cm²	12 cm²	●○○○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
32. For r = 5 cm and θ = 3 rad, area of sector =	39.5 cm²	15 cm²	75 cm²	37.5 cm²	○○○●
33. π/4 radians in degrees is	75°	90°	135°	45°	○○○●
34. Arc length for r = 6 cm and θ = π/3 is	3π cm	2π cm	6π cm	4π cm	○●○○
35. 20° in radians is	π/9	9π	4π/9	17π/36	●○○○
36. Angle between clock hands at 3 o'clock (radians)	2π	π/3	π	π/2	○○○●
37. For r = 10 cm and θ = 1 rad, area of sector =	25 cm²	100 cm²	50 cm²	10 cm²	○○●○
38. 240′ in degrees is	4°	8°	2.4°	14400°	●○○○
39. The angle 750° is coterminal with	840°	660°	570°	390°	○○○●
40. 7π/4 radians in degrees is	315°	630°	945°	345°	●○○○
41. 300″ in minutes is	18000′	0.08′	10′	5′	○○○●
42. 75° in minutes is	4500′	7500′	450′	75′	●○○○
43. 36° in radians is	13π/20	π/10	π/5	5π	○○●○
44. 75.2° in D°M′S″ form is	75°22′0″	75°12′30″	75°20′0″	75°12′0″	○○○●
45. The angle −45° is coterminal with	45°	135°	315°	405°	○○●○
46. 30°30′ in decimal degrees is	60°	30.5°	30.3°	35°	○●○○
47. The angle through which the hour hand turns in 1 hour is	6°	12°	30°	60°	○○●○
48. Sector area for r = 4 cm and θ = π/2 is	2π cm²	8π cm²	4π cm²	5π cm²	○○●○

StudyBees OFFICIAL PAK

Questions	A	B	C	D	A B C D
49. Sector area for r = 9 cm and θ = π/3 is	27π cm²	27π/2 cm²	27π/4 cm²	29π/2 cm²	○●○○
50. 60° in radians is	π/3	3π	2π/3	4π/3	●○○○
51. 30.75° in D°M′S″ form is	29°45′0″	30°55′0″	30°45′30″	30°45′0″	○○○●
52. The terminal ray of 720° lies along	Negative x-axis	Positive x-axis	Positive y-axis	Negative y-axis	○●○○
53. 240° in radians is	3π/2	16π/3	4π/3	35π/12	○○●○
54. 4π/9 radians in degrees is	160°	80°	110°	40°	○●○○
55. π/10 radians in degrees is	36°	18°	48°	54°	○●○○
56. The angle through which the minute hand turns in 1 minute is	60°	12°	30°	6°	○○○●
57. 45′ in seconds is	270″	162000″	2700″	45″	○○●○
58. 270° in radians is	5π/3	3π/2	3π/4	13π/4	○●○○
59. Sector area for r = 3 cm and θ = π/6 is	3π/2 cm²	3π/4 cm²	3π/8 cm²	π/2 cm²	○●○○
60. 30° in minutes is	108000′	180′	30′	1800′	○○○●
61. 180′ in seconds is	648000″	180″	10800″	1080″	○○●○
62. π/3 radians in degrees is	60°	180°	90°	120°	●○○○
63. The terminal ray of 540° lies along	Positive x-axis	Positive y-axis	Negative y-axis	Negative x-axis	○○○●
64. The coordinates of the point on the unit circle for 45° are	(√3/2, 1/2)	(1/2, √3/2)	(1, 1)	(1/√2, 1/√2)	○○○●
65. The coordinates of the point on the unit circle for 60° are	(√3/2, 1/2)	(1/√2, 1/√2)	(1/2, √3/2)	(0, 1)	○○●○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
66. The coordinates of the point on the unit circle for 30° are	(√3/2, 1/2)	(1/2, √3/2)	(1/√2, 1/√2)	(1, 0)	●○○○
67. In the third quadrant, which ratios are positive?	sin and cosec	tan and cot	cos and sec	All of them	○●○○
68. In the first quadrant, all trigonometric ratios are	Positive	Negative	Zero	Undefined	●○○○
69. In a right triangle, tan θ is	Base/Perpendicular	Perpendicular/Base	Perpendicular/Hypotenuse	Hypotenuse/Base	○●○○
70. sec θ is the reciprocal of	sin θ	tan θ	cot θ	cos θ	○○○●
71. In a right triangle, cos θ is	Base/Hypotenuse	Perpendicular/Hypotenuse	Perpendicular/Base	Base/Perpendicular	●○○○
72. On the unit circle, tan θ equals	y/x	x/y	xy	x + y	●○○○
73. In the fourth quadrant, which ratios are positive?	sin and cosec	tan and cot	cos and sec	All of them	○○●○
74. Which trigonometric ratio is undefined at 0°?	sec θ	cosec θ	tan θ	sin θ	○●○○
75. In a right triangle, sin θ is	Base/Hypotenuse	Perpendicular/Hypotenuse	Perpendicular/Base	Hypotenuse/Base	○●○○
76. The angle between 270° and 360° lies in the	Fourth quadrant	Third quadrant	Second quadrant	First quadrant	●○○○
77. Which trigonometric ratio is undefined at 180°?	cosec θ	sec θ	tan θ	cos θ	●○○○
78. In the second quadrant, which ratios are positive?	cos and sec	tan and cot	All of them	sin and cosec	○○○●
79. The angle between 180° and 270° lies in the	Third quadrant	Second quadrant	Fourth quadrant	First quadrant	●○○○
80. Which trigonometric ratio is undefined at 270°?	sin θ	tan θ	cosec θ	cot θ	○●○○
81. cos 60° =	2	1/√3	√3/2	1/2	○○○●
82. sin 60° − cos 30° =	1/2	√3	0	1	○○●○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
83. cot 90° =	1	1/2	0	∞	○○●○
84. sin 60° =	2	√3/2	1/2	2/√3	○●○○
85. sin²60° + cos²60° =	3/4	2	1/2	1	○○○●
86. If tan θ = −1 and θ is in the fourth quadrant, sin θ =	1	−1/√2	1/√2	−√3/2	○●○○
87. Sign of sin θ in the third quadrant	Positive	Negative	Zero	Undefined	○●○○
88. sec 45° · cosec 45° =	2	√2	4	1/2	●○○○
89. cos 0° =	1	2	1/√2	0	●○○○
90. cos 270° =	−1	∞	0	2/√3	○○●○
91. tan 45° + cot 45° =	√2	2	√3	1	○●○○
92. If tan θ = 5/12 (θ acute), sec θ =	13/12	5/13	12/5	13/5	●○○○
93. tan²60° =	1/3	3	9	1	○●○○
94. Sign of cot θ in the second quadrant	Positive	Zero	Undefined	Negative	○○○●
95. sin 180° + cos 180° =	0	2	−1	−2	○○●○
96. tan θ = sin θ / cos θ is a	Reciprocal identity	Pythagorean identity	Sum identity	Quotient identity	○○○●
97. (x + 1)(x − 1) = x² − 1 is an example of an	Equation	Identity	Inequality	Radical	○●○○
98. (1 − sin²θ) sec²θ equals	0	1	sin²θ	tan²θ	○●○○
99. The identities cosec θ = 1/sin θ etc. are called	Quotient identities	Pythagorean identities	Double angle identities	Reciprocal identities	○○○●

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
100. 1 + cot²θ equals	cosec²θ	sec²θ	tan²θ	cos²θ	●○○○
101. 1 − sin²θ equals	sin²θ	tan²θ	sec²θ	cos²θ	○○○●
102. cot θ equals	sin θ / cos θ	cos θ / sin θ	sin θ cos θ	1/cos θ	○●○○
103. 1 + tan²θ equals	cosec²θ	cot²θ	sin²θ	sec²θ	○○○●
104. (1 − cos²θ) sec²θ equals	1	cot²θ	tan²θ	sec²θ	○○●○
105. The identity cosec²θ − cot²θ equals	0	cos²θ	sec²θ	1	○○○●
106. An equation true for all values of its variable is called an	Identity	Inequality	Equation only	Expression	●○○○
107. sin θ cosec θ =	1	sin²θ	tan θ	−1	●○○○
108. In right ΔABC, ∠C = 90°, a = 5, c = 13. Side b =	18	12	8	√194	○●○○
109. tan θ cot θ =	sin θ	1	0	tan²θ	○●○○
110. cos θ / sin θ =	cot θ	tan θ	sec θ	cosec θ	●○○○
111. (1 − sin²θ) / cos²θ =	sec²θ	0	sin²θ	1	○○○●
112. A right isosceles triangle with legs 10 cm has hypotenuse	100 cm	10√2 cm	20 cm	5√2 cm	○●○○
113. cos θ sec θ =	1	−1	0	cos²θ	●○○○
114. In right ΔPQR, ∠Q = 90°, ∠P = 60°, PQ = 4√3 cm. PR =	8 cm	12 cm	4√3 cm	8√3 cm	○○○●
115. Sum of acute angles of a right triangle is	45°	90°	180°	60°	○●○○
116. sec θ − cos θ =	sin θ cot θ	cos θ cot θ	sin θ tan θ	cos θ tan θ	○○●○

StudyBees OFFICIAL PAK

Questions	A	B	C	D	A B C D
117. The angle between the horizontal and the line of sight to an object below it is the angle of	Elevation	Rotation	Bearing	Depression	○○○●
118. If the bearing of B from A is α (acute), the bearing of A from B is	180° − α	180° + α	360° − α	α	○●○○
119. Bearings are measured from	East	South	West	North	○○○●
120. The bearing of south-west is	045°	135°	225°	315°	○○●○
121. Intercardinal directions include	North	East	South	North-east	○○○●
122. The angle between the horizontal and the line of sight to an object above it is the angle of	Depression	Rotation	Elevation	Bearing	○○●○
123. The bearing of north-east is	135°	225°	315°	045°	○○○●
124. The bearing of west is	270°	180°	090°	360°	●○○○
125. The line joining the observer to an object is the	Horizontal line	Vertical line	Line of sight	Radius	○○●○
126. Bearings are measured in the	Anti-clockwise direction	Clockwise direction	Vertical direction	Radial direction	○●○○
127. Bearings are written as	Two-digit numbers	Fractions	Three-digit numbers	Radians	○○●○
128. The bearing of south is	090°	180°	270°	000°	○●○○
129. Mostly bearings are used by	Sailors and pilots	Bankers	Cooks	Painters	●○○○
130. The bearing of east is	000°	090°	180°	270°	○●○○
131. Angles of depression and elevation (alternate angles) are	Supplementary	Complementary	Equal	Unrelated	○○●○
132. Exact north has a bearing of	0°	90°	180°	360° only	●○○○
133. The bearing of north is	000°	090°	180°	270°	●○○○

Follow Studybeesofficial on Instagram, YouTube Channel & WhatsApp Channel — Link in Bio

Questions	A	B	C	D	A B C D
134. The bearing of north-west is	045°	135°	315°	225°	○○●○
135. If the bearing of B from A is 165°, the bearing of A from B is	075°	345°	255°	165°	○●○○
136. Angle of elevation of a tree top from 30 m away is 60°. Height =	30/√3 m	30 m	30√3 m	60 m	○○●○
137. A car travels 13 km at bearing 040°. Distance travelled east ≈	9.96 km	8.36 km	4.64 km	13 km	○●○○
138. Hot air balloon 500 m away rises so elevation changes 30° to 60° in 100 s. Speed ≈	2.89 m/s	11.5 m/s	5.77 m/s	8.66 m/s	○○●○
139. Angle of elevation of a tree top from 45 m away is 60°. Height =	45/√3 m	45 m	45√3 m	90 m	○○●○
140. If the bearing of B from A is 310°, the bearing of A from B is	130°	050°	040°	310°	●○○○
141. A 10 m pole has shadow 10√3 m. Angle of elevation of the sun is	90°	15°	30°	45°	○○●○
142. If the bearing of B from A is 290°, the bearing of A from B is	020°	110°	200°	290°	○●○○
143. Angle of elevation of a tree top from 20 m away is 30°. Height =	20/√3 m	20√3 m	20 m	40/√3 m	●○○○
144. A 10 m pole has shadow 10 m. Angle of elevation of the sun is	45°	15°	60°	30°	●○○○
145. A car travels 50 km at bearing 045°. Distance travelled east ≈	50 km	14.64 km	35.36 km	36.36 km	○○●○
146. If the bearing of B from A is 060°, the bearing of A from B is	060°	150°	240°	300°	○○●○
147. From height 100 m, depression to bottom of another building is 30°. Distance between buildings =	200 m	100/√3 m	50 m	100√3 m	○○○●
148. If the bearing of C from B is 180° and C lies east of A with AB = 10 km, ∠BAC = 45°, then AC ≈	20 km	14.14 km	5 km	7.07 km	○○○●
149. A 10 m pole has shadow 10/√3 m. Angle of elevation of the sun is	30°	15°	60°	45°	○○●○
150. A car travels 20 km at bearing 060°. Distance travelled north ≈	10 km	17.32 km	20 km	11 km	●○○○